SOLARSAFE

Materials Team

Material selection for safer maize drying containers

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Overview: Why Safe Maize Matters

People in Kenya and across Africa eat maize-based foods such as **Nsima, Sadza, etc.**

Aflatoxin-contaminated maize

People

- Acute poisoning
- Liver damage
- Higher liver cancer risk
- Child growth and immune effects

Animals

- Reduced animal health
- Meat and milk can be affected
- Food-chain safety concerns

How can we stop aflatoxin before people and animals are exposed?

The Problem: Aflatoxin risk starts before storage

SOLARSAFE dries maize in step 2 to prevent aflatoxin production.

1 Harvest

2 Enclosed Solar Dryer

3 Moisture Check

4 Storage

Goal: Find optimized bag material for maize drying.

SOLARSAFE Solution Overview

SOLARSAFE goal

Make maize drying safer, more controlled, and more reusable.

Materials Team - main question

Which plastic is most suitable for a reusable maize drying bag?

Evaluation criteria

- Resistance to sodium hypochlorite
- Surface stability after exposure
- Transparency
- Cost
- Future oxygen and moisture performance

Stakeholders

Farmers

Less post-harvest loss

Healthcare system

Reduced long-term food safety burden

Government / food system

Better food security

Technical Section

SOLARSAFE Solutions Overview

1. Sodium Hypochlorite-surface interaction - Experiments

Key Question: Does sodium hypochlorite degrade the drying plastic bag?

2. Plastic moisture trapping level -Experiment

Key Question: Which plastic maintains the highest moisture level inside?

Initial Materials Selection

Shortlisted for testing

✓ **PP** Polypropylene	Flexible, low cost.
✓ **PVC** Polyvinyl chloride	Used by Benard's group currently, strong and transparent.
✓ **Nylon** Polyamide	More tough than PE, less brittle than PET.

For Future Exploration

PE (HDPE) Polyethylene	Low water permeability, would trap the evaporated moisture.
LDPE Low-density PE	Through lit review: too good at trapping aflatoxin.
PET Polyester	Low water permeability, would trap the evaporated moisture. Not very flexible.

Experiment 1: External Degradation

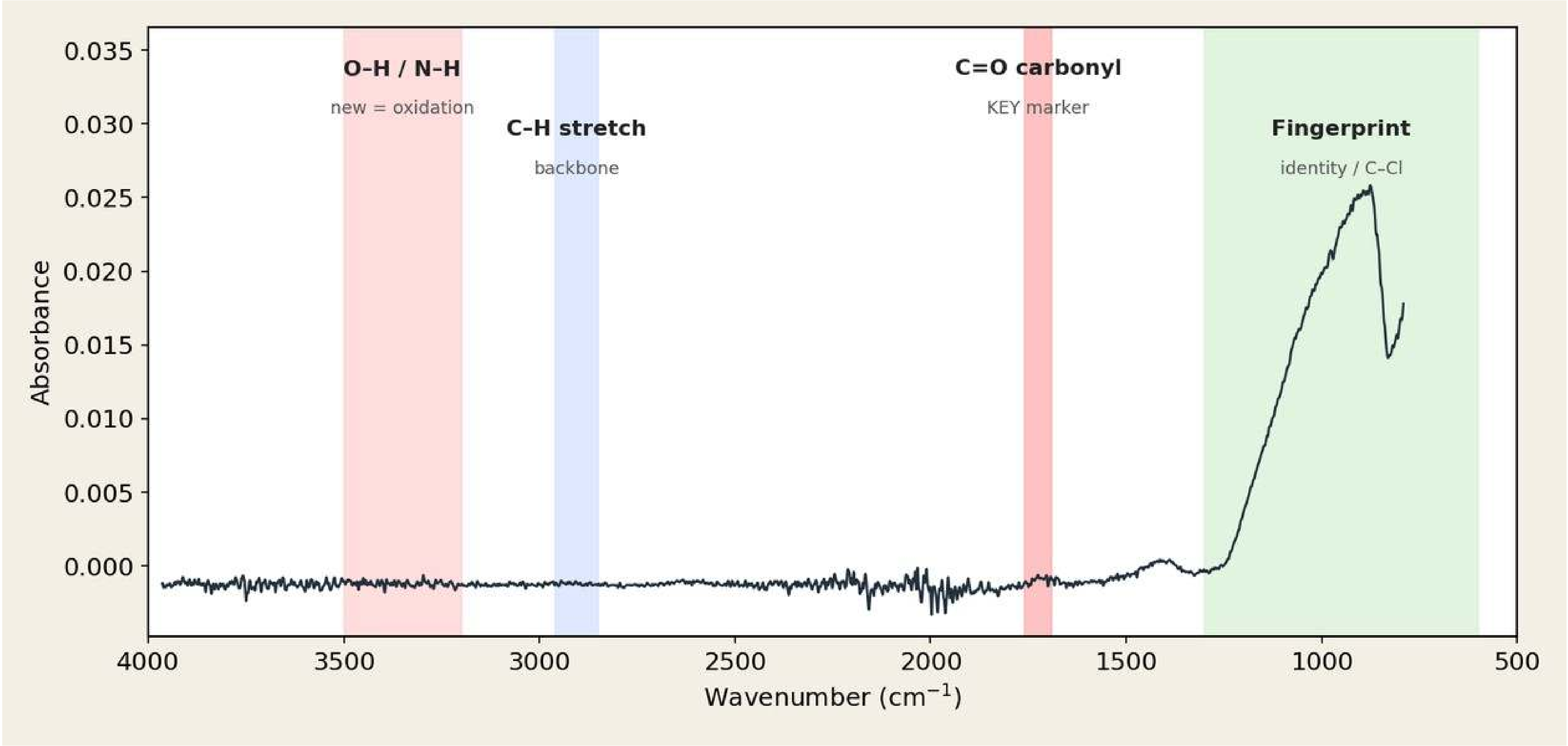
1)	**OBJECTIVE** **Evaluate material resistance to sodium hypochlorite**
2)	**METHOD** PVC, PP and Nylon exposed to: • Water (control) • 3% NaOCl • 5% NaOCl • 5% NaOCl + heat
3)	**ANALYSIS** ATR-FTIR Optical Microscopy (Profilometry unavailable)
4)	**AIM** **Identify materials that can withstand cleaning and long-term use**

Experiment 1 results: Plastic Degration in Sodium hypochlorite, Analyzed by ATR

How to read a spectrum

- The two **RED zones** are 'damage zones' (O−H ~3300 and C=O ~1715). **The bumps here indicate plastic degradation.**
- The **GREEN zone** is the plastic's normal fingerprint (its identity). **Those bumps are meant to be there.**

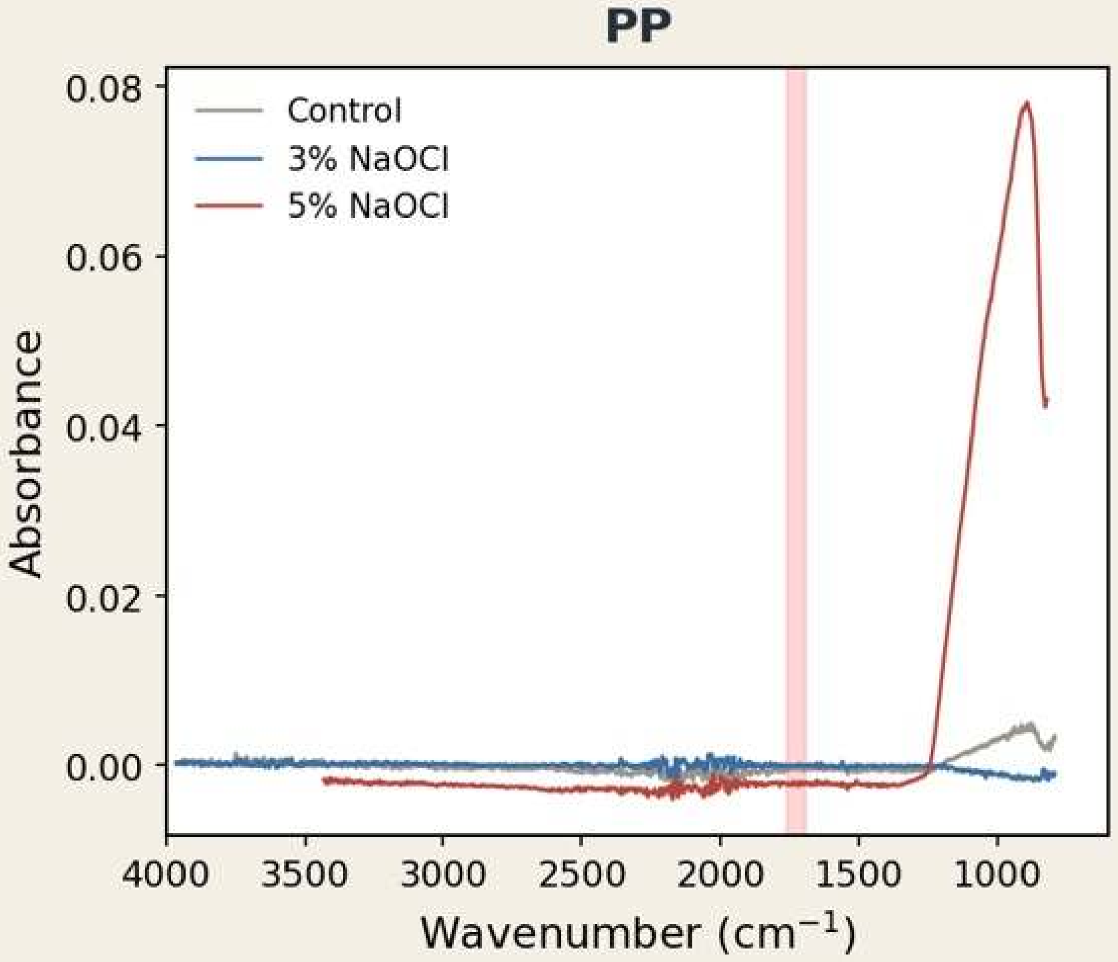
- Warning sign: a green-zone fingerprint bump that shrinks, vanishes or shifts = the plastic is breaking down.

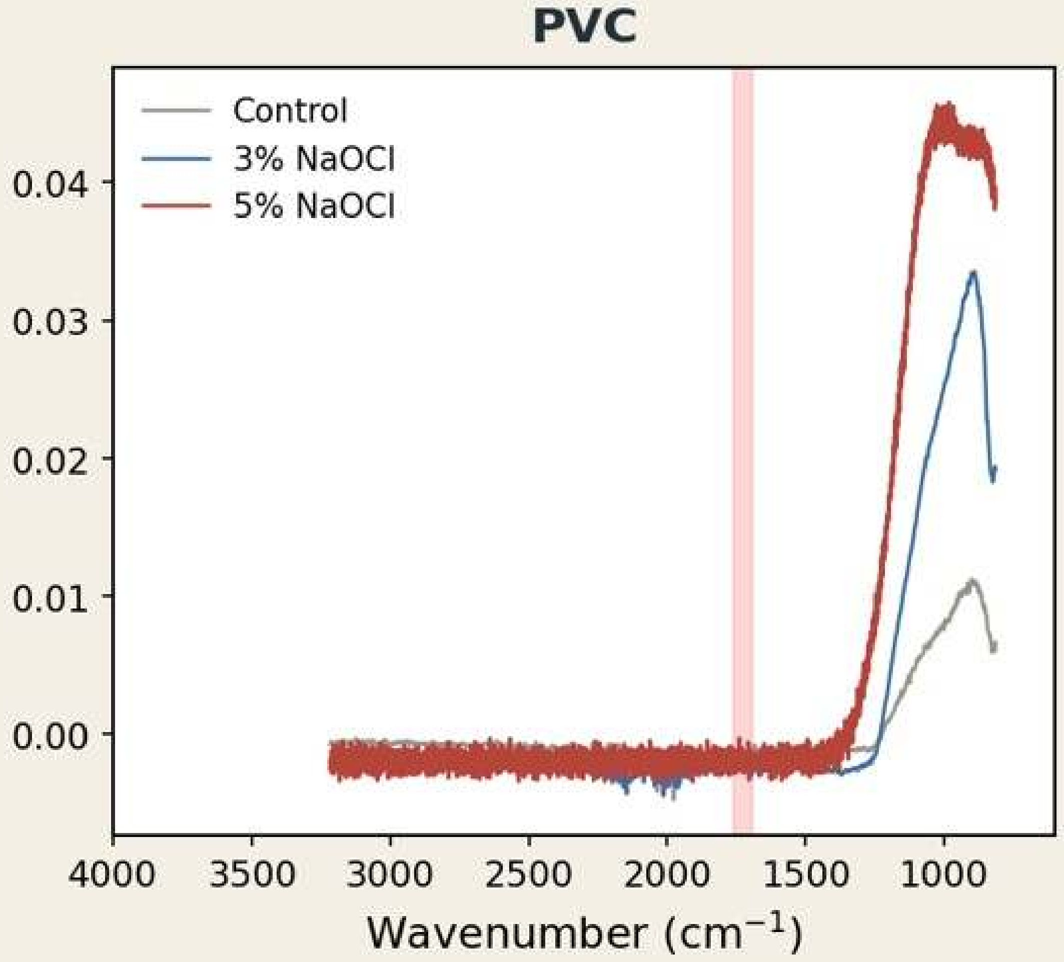


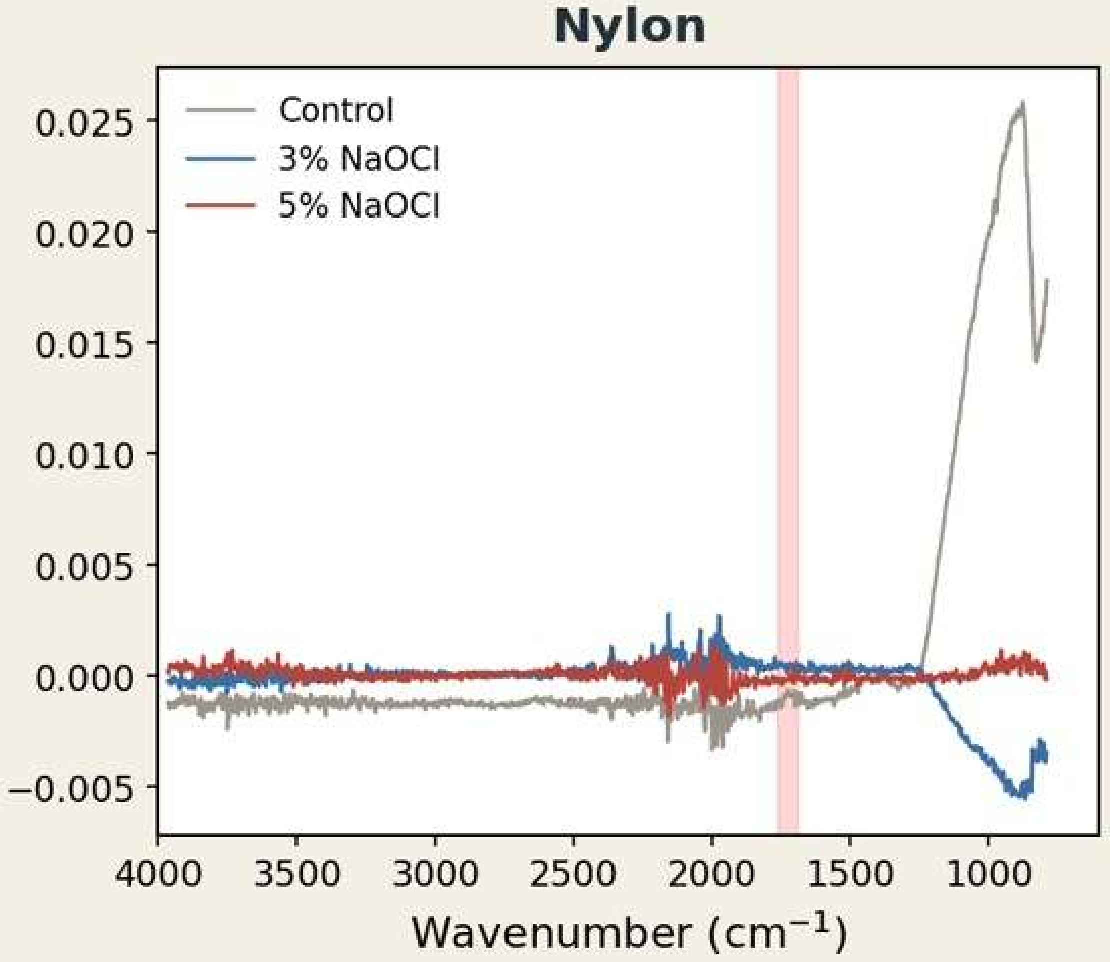
Rule of thumb: spectrum unchanged → material is stable in bleach. New carbonyl / hydroxyl bands → oxidation and degradation.

Results: PP, PVC & Nylon after 3 Days in NaOCl

What the spectra show







PP

No new bumps in red zone. Backbone unchanged, no sign of attack.

PVC

No new bumps in red zone. Stable, no breakdown seen.

Nylon

Signal weak and noisy, slight 3% drift only in red zone. Further experiments should be conducted.

Take-away: No evidence of degradation after 3 days, PP and PVC look compatible with NaOCl cleaning, while Nylon needs more testing.

Experiment 2: Internal Conditions

Team

day 0

- experimental set up
- seal plastic bags with corn and sensors inside
- control corn outside
- CO2, humidity nd temp

Kunran

day 2

- measurements
- ave measurements with sensors
- Moisture probe measuremnet in corn (instead of John Deere tube)

Mirha

day 3

- unseal bags due replication of opening/closing drying container
- also prevent sensor saturation

day 4

24 H

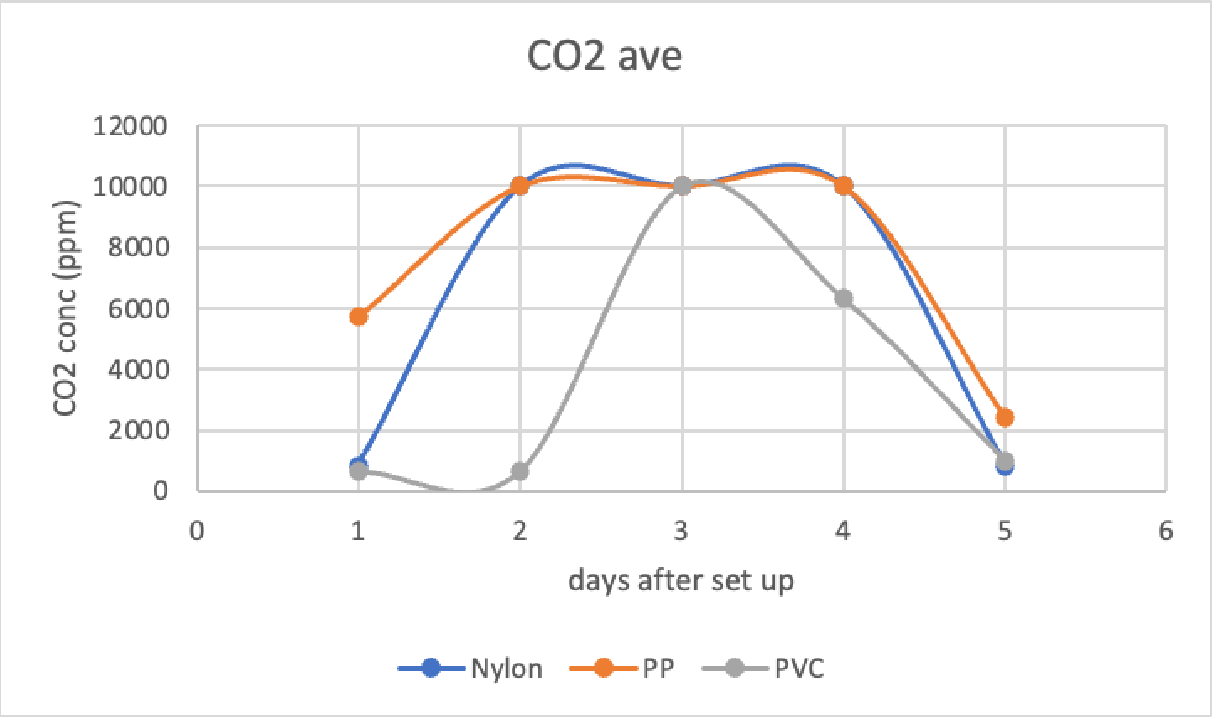
Mirha **Kunran**

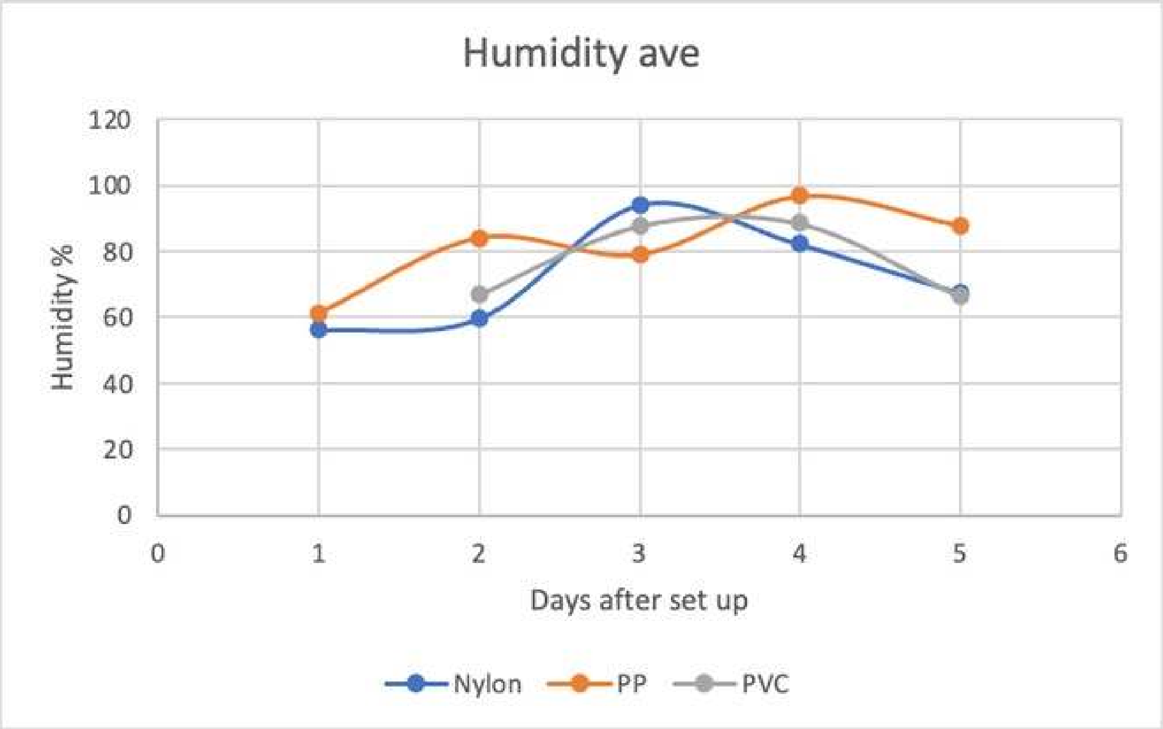
Final Analysis

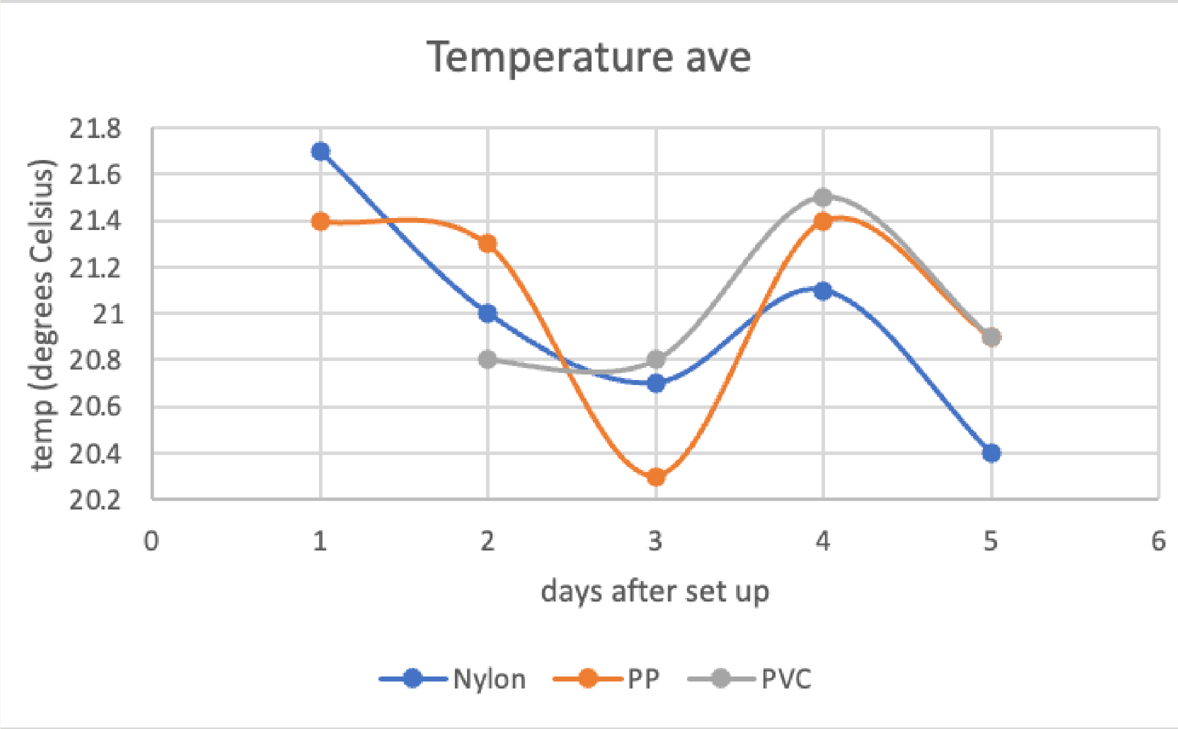
anaylsis and results

- Co2 rise, then fall after unseal
- temp very small variations hard to tell whether significant, rise with Co2
- moisture rises
- moisture falls for control

Experiment 2 Results







CO_2

- Increased in all bags by days 2–3.
- Nylon and PP reached sensor limit first.
- PVC increased more slowly.
- Decreased after opening on day 3.
- PP had highest CO_2 on day 5.

Moisture (ave)

- Increased while bags were sealed.
- Changed after opening on day 3.
- Differences observed between plastics.
- Indicates different moisture retention properties.

Temperature

- Little overall variation.
- Nylon highest on day 1.
- PP lowest on day 3.
- PVC highest on day 4.
- No clear link to CO_2.

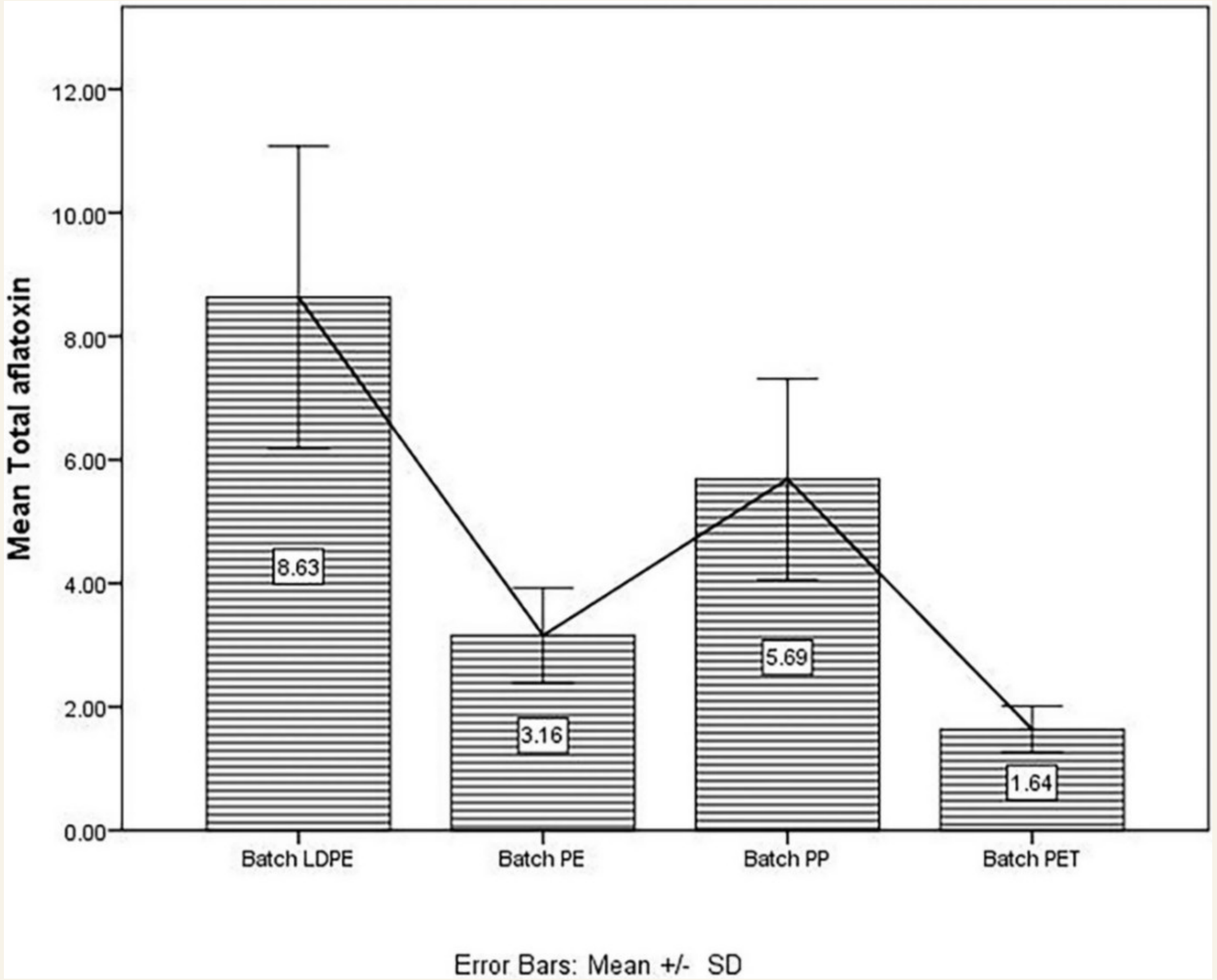
Literature Review: Aspergillus

Key Finding : Aspergillus degrades LDPE severely, so we filtered out in our material selection

- **Samples:** LDPE 15 samples, PE 15 samples, PP 15 samples, and PET 19 samples. 64 ready-packed products: nuts and dried fruits stored in plastic packaging.

- **Total aflatoxin levels:** were highest in LDPE, followed by PP, then PE, and lowest in PET.

- **Material Selection:** If time allowed, **we would also test PE and PET** as literature comparators.



Mean concentration of total aflatoxins (μg/kg) for each batch of screened samples. (ELISA-based method)

Source: Macri et al., The Occurrence of Aflatoxins in Nuts and Dry Nuts Packed in Four Different Plastic Packaging from the Romanian Market, Microorganisms, 2021.

Final Material Recommendation

Recommended material:

Best balance of chemical resistance, integrity, safety and reuse for the drying bag.

How each plastic performed	PP	PVC	Nylon
Resists sodium hypochlorite (FTIR, 3 days)	✓ No change	✓ No change	~ Unclear
Keeps material integrity	✓ Stable	~ Can turn brittle	~ Absorbs water
Moisture	retains the highest moisture	retained least	intermediate
CO_2 (O_2) permeability	best barrier, CO_2 accumulation	intermediate, relatively low	lowest accumulation of CO_2
Lightweight, low-cost, available	✓ Yes	~ Moderate	✗ Heavier/costly

✓ good ~ fair ✗ poor

Recommendations: PVC has the greatest ability to withstand chemical stress of disinfection. Nylon has the greatest permeability to CO_2 (and thus likely O_2) which prevents the accumulation of CO_2, prevents hot spot formation. PP degrades most under long term simulated stress and also has worst permeability leads to accumulation of CO_2 and moisture..

What We Personally Contributed

Kunran

- Defined materials selection logic
- Led literature review on aflatoxin / plastic interaction
- Helped design humidity exposure experiment
- Contacted academics and labs
- Supported measurements and result interpretation
- Developed final comparison and recommendation

Mirha

- Coordinated lab preparation and risk assessment
- Helped design NaOCl exposure experiment
- Prepared materials and sample setup
- Contacted academics and labs
- Supported measurements and result interpretation
- Developed final comparison and recommendation

Limitations and Next Steps

Limitations

- Short experiment time scale
- Lab exposure may not fully represent field conditions
- Real maize drying involves variable weather (sunlight, rain, dust)
- didn't have access to best equipment/techniques like ELISA assays or John Deere tubes

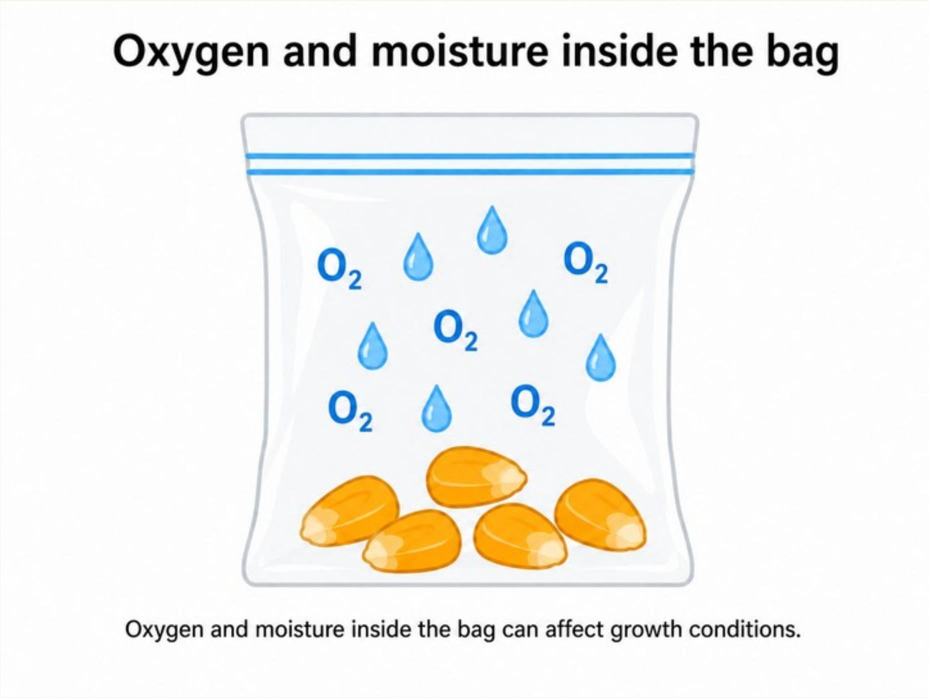
Next steps

- Longer ageing test
- Test with a profilometer to analyze chemical changes on the surface
- Field test in Kenya with SOLARSAFE dryer design
- Assays experiments

Future Challenges to Explore

1. Altering materials

- Non flexible materials = durability enhanced?
- No dark under surface = prevent temperature gradient to prevent localised regions of heat that might be conducive to fungal growth



Oxygen and moisture inside the bag can affect growth conditions.

2. Aflasafe (natural biocontrol product) for Kenya

Can we develop a biocontrol product for Kenya to **prevent pre-harvest corn infection**?

Final Conclusion

Our experiments turn SOLARSAFE from a concept into a testable design decision. Based on the results, we recommend **PVC as the most suitable material** for the prototype because it showed strong durability during cleaning, maintained stable drying performance, and is practical and affordable for agricultural use.

Thank You!